Measuring What Matters in Security Operations Centers:
A Proposed Goal-Coherence Framework for Security Metrics

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Agency Theory · Strategy Coherence · Mixed-Methods Evidence · Goal-Metric Mapping Framework

BACKGROUND

SOCs spend millions on metrics that reward operational busyness rather than actual risk reduction.

- *KPIs such as MTTR and ticket volume measure visible activity.*
- *These indicators do not always reflect strategic security value.*
- *Result: dashboards can look healthy while risk remains*

Grounded in Agency Theory, this study maps goal incongruence between CISO priorities and analyst KPIs using an academic SOC case study, LLM-assisted task extraction, and ongoing expert validation.

RESEARCH PROBLEM

The measurement paradox: CISOs are accountable for security outcomes, but analysts are often evaluated using speed, throughput, and closure-based KPIs. When what is measured differs from what the organization truly values, analysts are incentivized to optimize the wrong behaviors.

RESEARCH QUESTION

How can SOC performance measurement be redesigned so that analyst incentives align more closely with strategic organizational cybersecurity goals rather than proxy indicators of operational busyness?

AGENCY THEORY AND GOAL INCONGRUENCE IN THE SOC

From Strategic Security Goals to Proxy Metric Drift

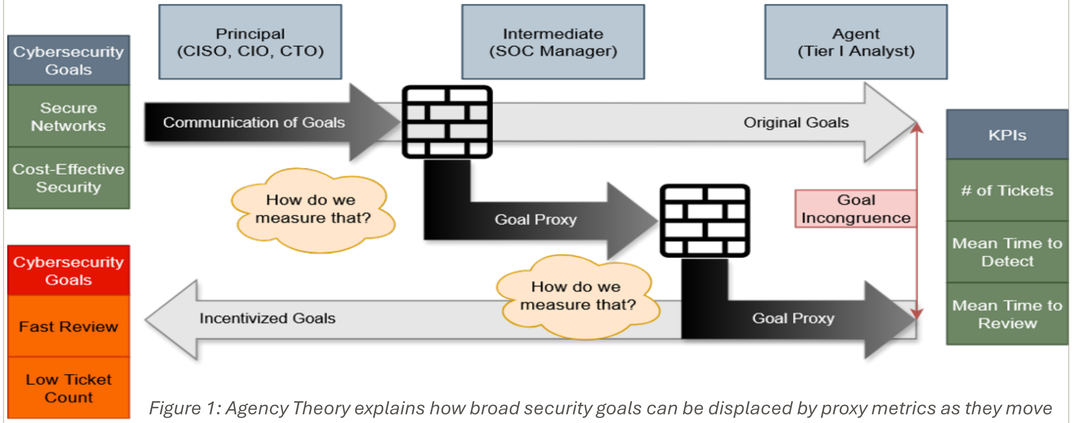

Figure 1: Agency Theory explains how broad security goals can be displaced by proxy metrics as they move through the SOC hierarchy, creating goal incongruence between executive intent and analyst incentives.

- CISOs cannot directly observe analyst security value
- Organizations default to behavior-based proxies (speed, volume)
- Analysts rationally optimize what is measured — not what matters

  "Simple terms: The boss wants security. The analyst is rewarded for clearing the inbox."

RESEARCH OBJECTIVE

To develop a Goal-Metric Mapping Framework that links analyst tasks to strategic SOC goals and makes measurement misalignment visible, explainable, and actionable for organizational decision-makers.

MIXED-METHODS DESIGN

From Empirical Observation to Framework Construction

1. **Academic SOC Case Study**
   Empirical comparison of KPI logic and perceived security value in a live SOC
2. **Industry Artifact Collection**
   49 job descriptions as leadership-facing representations of expected analyst work
3. **LLM-Assisted + Human-Validated Task Extraction**
   49 JDs → 211 unique SOC tasks and standardized thematic categories
4. **Task-to-Goal Mapping**
   Analyst activities mapped against strategic SOC goals
5. **Expert Validation**
   Ongoing Delphi-style validation to refine mappings and coherence scoring
6. **Framework Output**
   Preliminary Goal-Coherence Matrix and future Strategy Coherence Score

Figure 2: The research combines qualitative and computational analysis to move from observed measurement misalignment toward a validated framework for evaluating whether SOC metrics support or distort strategic intent.

CURRENT STATUS

- Misalignment empirically demonstrated in a live SOC
- 211 unique tasks extracted and categorized from 49 job descriptions
- Expert validation underway framework publication in preparation
- *Strategy Coherence Score formalization and broader field validation planned for completion by August 2026.*

GOAL-COHERENCE MATRIX

A diagnostic tool for SOC metric alignment: Illustrative examples of strategic alignment of common SOC metrics (preliminary / ongoing)
How to read this: Each row maps a common SOC metric to its hidden behavioral incentive and its true alignment with strategic security goals. Red = actively works against security intent. Amber = neutral or context-dependent. Green = supports strategic goals.

Task Context	Current Metric	Hidden Incentive	Coherence	Risk Signal
Security Monitoring & Alert Triage	Alert Processing Volume	Clear queue faster	Misaligned	High false confidence
Security Monitoring & Alert Triage	Mean Time to Detect (MTTD)	Detect quickly	Neutral	Context dependent
Incident Response & Containment	Tickets Closed per Shift	Close more cases	Misaligned	Important incidents may be rushed
Incident Response & Containment	Escalation Accuracy	Escalate correctly	Aligned	Better triage quality
Threat Analysis & Investigation	Quality of Analysis	Quality of Analysis	Aligned	Stronger decision support
Professional Development	Analyst Competency / NPS	Build expertise	Aligned	Improves long-term resilience

⚠ These metrics are in active use across most SOCs today

CONTRIBUTION AND SIGNIFICANCE

This research reframes SOC inefficiency as a measurement-design problem, not an analyst capability problem.

- *Empirical — In a live academic SOC, KPI-based analyst rankings showed near-zero correlation with leadership-assessed security value, suggesting metrics captured activity rather than strategic contribution.*
- *Theoretical — Agency Theory explains why proxy metrics persist even when leaders know they are imperfect*
- *Practical — The GCM gives leaders a diagnostic tool to audit, redesign, or retire misaligned KPIs without replacing existing infrastructure*

Takeaway: SOCs often measure what is easy to count rather than what reduces risk. This framework makes the difference visible.

KEY TERMS

SOC: Security Operations Center CISO: Chief Information Security Officer GCM: Goal-Coherence Matrix
KPI: Key Performance Indicator MTTR: Mean Time to Respond LLM: Large language Models
NPS : Net Promoter Score CTO: Chief Technology Officer CIO: Chief Information Officer

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Scan for extended artifact & Preliminary full matrix
***Validation ongoing.*
or visit ***https://bit.ly/4c9nm1u***